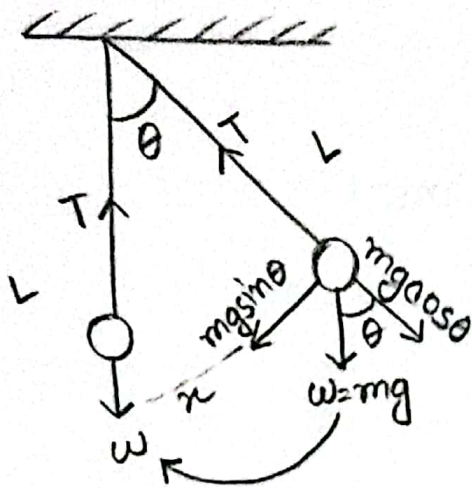


θ মেদিকে কোণ টেনশন করে যে
বরাবর $\cos\theta$ হবে।

Tension Force: Tension is defined as the force transmitted through a rope, string or wire when pulled by forces acting from opposite sides.

Tension always thread (অঁত) বরাবর কাজ করবে।

Tension force এবং $mg \cos\theta$ একে অপরের সম্পূর্ণ বিপরীত দিকে কাজ করে, তাই vanish হয়ে যাবে। সুতরাং $mg \sin\theta$ থাকবে,

$T = mg \cos\theta$ বিপরীত হয় অন্যত।

$$\text{Restoring} = -mg \sin\theta$$

$$= -mg\theta$$

$$x = L\theta$$

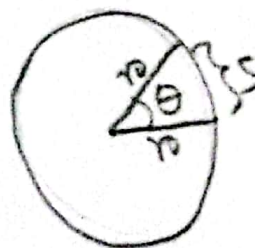
$$\Rightarrow \frac{d^2x}{dt^2} = L \frac{d^2\theta}{dt^2} \quad [\text{a এর করার জন্য দুইবার differentiation}]$$

$$\Rightarrow a = L \frac{d^2\theta}{dt^2}$$

θ এর মান যদি অনেক ছোট হয় তখন $\sin\theta \approx \theta$

$\theta = \text{angular displacement}$

$x = \text{linear displacement}$



$$\therefore s = r\theta$$

$$\therefore x = L\theta$$

We know,

$$F = -mg \sin \theta$$

$$\Rightarrow ma = -mg \sin \theta$$

$$\Rightarrow ma = -mg \theta$$

$$\Rightarrow a = -g \theta$$

$$\Rightarrow L \frac{d^2 \theta}{dt^2} = -g \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\frac{g}{L} \theta$$

$$\Rightarrow \frac{d^2 \theta}{dt^2} = -\omega^2 \theta$$

$$\Rightarrow \boxed{\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0}$$

now, $\omega^2 = \frac{g}{L}$

$$\Rightarrow \omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{L}{g}}$$

(proved)

in spring $\omega^2 = \frac{k}{m}$

in angle, $\omega^2 = \frac{g}{L}$

in angular displacement,

$$\frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

in linear displacement,

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

Simple Pendulum

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Math-1: $T_1 = 2s$, $g_1 = 9.78 \text{ ms}^{-2}$

$T_2 = 3s$, $g_2 = ?$ [L is constant in here]

$$T_1 = 2\pi\sqrt{\frac{L}{g}} \text{ ——— ①}$$

$$T_2 = 2\pi\sqrt{\frac{L}{g}} \text{ ——— ②}$$

$$\text{①} \div \text{②}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}}$$

$$\Rightarrow g_2 = \frac{(T_1)^2}{(T_2)^2} g_1$$

• Theory Question: Simple pendulum satisfies the differential equation of simple harmonic motion.

• Determination of the spring constant and effective mass of a given spiral spring.

Math: 2

$$L_1 = L$$

$$L_2 = 2L$$

$$T_1 = T$$

$$T_2 = ?$$

• next length will be doubled up of previous length.

(g is constant in here cause the location didn't change)

$$T_1 = \sqrt{\frac{L_1}{g}}, \quad T_2 = \sqrt{\frac{L_2}{g}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$$

$$\frac{T_2}{T_1} = \sqrt{\frac{L_2}{L_1}}$$

$$\Rightarrow T_2 = \sqrt{\frac{L_2}{L_1}} \times T_1$$

$$\Rightarrow T_2 = \sqrt{\frac{2L}{L}} \times T_1$$

$$= T_2 = \sqrt{2} \times T_1$$

(LC Oscillations)

LC circuit: consists of inductor and capacitor

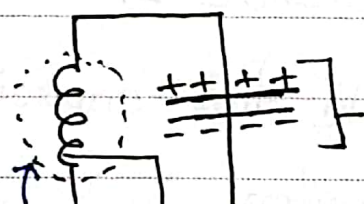
inductor

current

passing

magnetic field

inductor



concept of magnetic force →

current flow and vice versa)

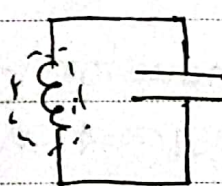
conservation of energy

capacitor: can store charge (current flow)

Flow of

change:

SHM

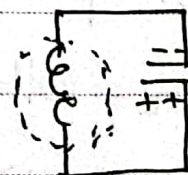


(current left)

no current flow

New Magnetic field is

VESTAR MR creating current
flow again



change of flow pole

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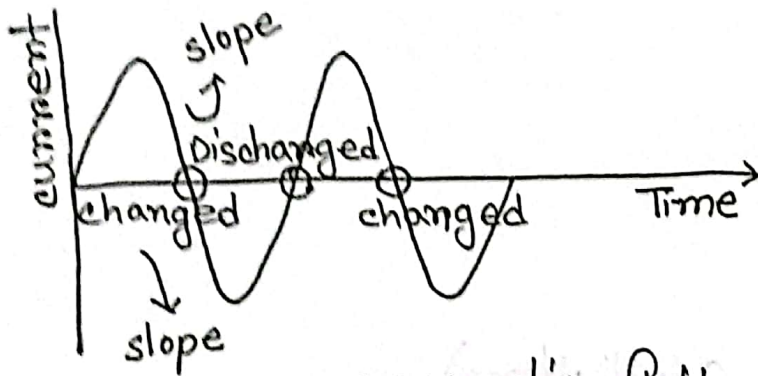
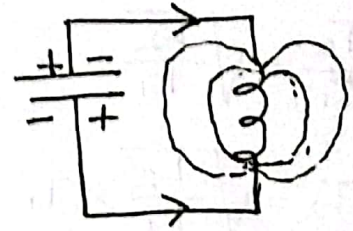
capacitor

* Circuit a magnetic field create and capacitor charge change and reverse polarity change and reverse direction.

RC circuit:



→ This pattern is called as LC pattern



magnetic field current

* First time capacitor discharged and create magnetic field and create reverse EMF

$Q = CV_c$ → Plate voltage $Q = CV_c$ for capacitor

$$V_c = \frac{Q}{C}$$

V_c = voltage of capacitor

C = capacitor

Q = plate voltage

Capacitor

$$V_L = L \frac{di}{dt}$$

V_L = voltage of inductor [inductor and current

i = current flow

t = time

প্রান্তের voltage]

$$V_L + V_C = 0$$

$$\Rightarrow L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d}{dt} \left(\frac{dQ}{dt} \right) + \frac{Q}{C} = 0 \quad \left[i = \frac{dQ}{dt} \right]$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + \frac{Q}{C} = 0$$

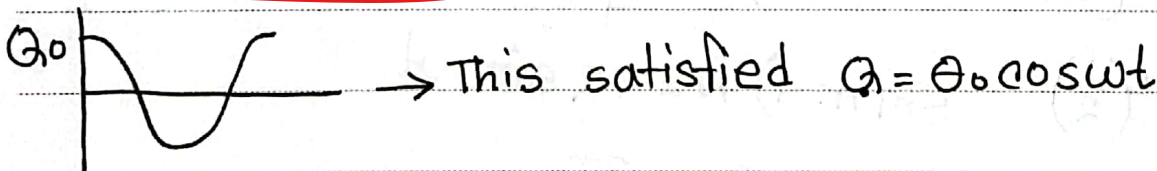
$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{1}{LC} Q = 0 \quad \left[\omega_0^2 = \frac{1}{LC} \right]$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \omega_0^2 Q = 0$$

what can be the possible solutions -

$$Q = Q_0 \sin \omega t$$

$$Q = Q_0 \cos \omega t$$

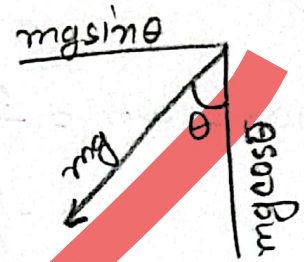
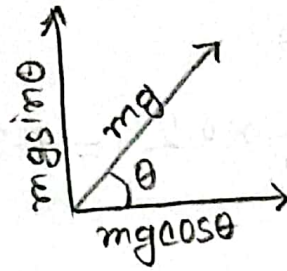
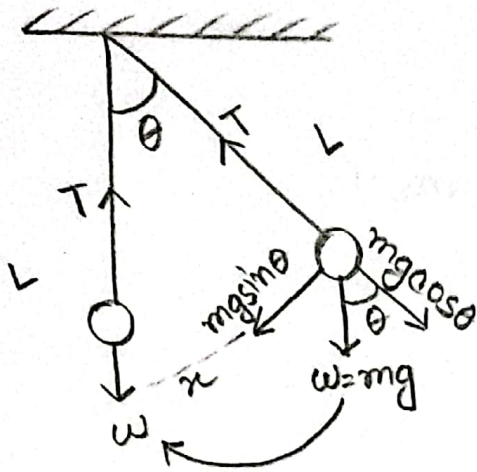


Can you calculate the current flow?

$$I = \frac{dQ}{dt} = \frac{d}{dt} (Q_0 \cos \omega t)$$

$$= -\omega Q_0 \sin \omega t$$

$$I = -I_0 \sin \omega t$$



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$T = mg \cos\theta$ বিপরীত হয় অন্য।

$$\text{Freestone} = -mg \sin\theta$$

$$= -mg\theta$$

θ এর মান যদি অনেক
ছোট হয় তাহলে $\sin\theta \approx \theta$

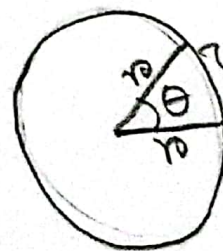
θ = angular displacement

x = linear displacement

$$x = L\theta$$

$$\Rightarrow \frac{d^2x}{dt^2} = L \frac{d^2\theta}{dt^2} \quad [\text{এর কারণ
জন্য দুইবার
differentiation}]$$

$$\Rightarrow a = L \frac{d^2\theta}{dt^2}$$



$$\therefore s = r\theta$$

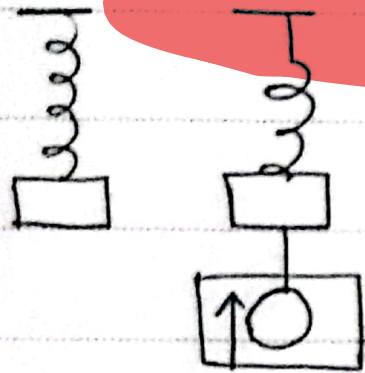
$$\therefore x = L\theta$$

$$E_{p(5)} = 980 \sin^2 \left(10 \times 5 - \frac{\pi}{6} \right)$$

$$= 493.6 \text{ J}$$

physics, class - 08 09

Damped Oscillation



$$F \propto -v$$

$$F' = -bv \rightarrow \text{constant}$$

$$ma = -kx - bv$$

$$\Rightarrow m \frac{d^2x}{dt^2} = -kx - b \frac{dx}{dt}$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\overset{\omega^2}{\left[\frac{k}{m} \right]} x - \left[\frac{b}{m} \right] \frac{dx}{dt} \quad \text{let } \gamma$$

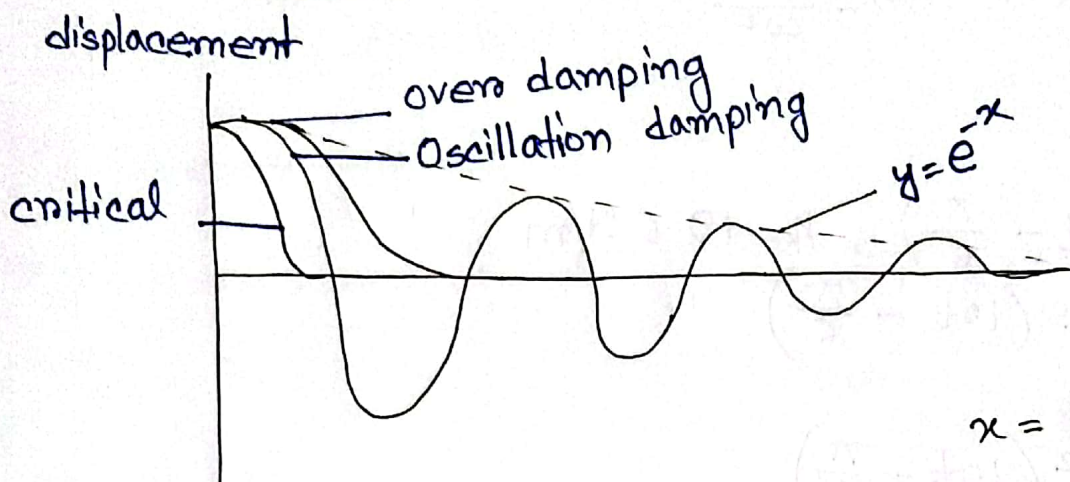
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$$\Rightarrow \frac{d^2x}{dt^2} = -\omega^2 x - \gamma \frac{dx}{dt} \quad \therefore \omega^2 = \frac{k}{m}, \quad \gamma = \frac{b}{m}$$

$$\Rightarrow \frac{d^2x}{dt^2} + \boxed{\gamma \frac{dx}{dt}} + \omega^2 x = 0 \quad \begin{array}{l} \text{came from} \\ \text{damping} \end{array}$$



$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

$$\Rightarrow \boxed{Bp^2 e^{pt}} + \gamma \boxed{Bpe^{pt}} + \omega_0^2 B e^{pt} = 0$$

$$\Rightarrow p^2 + \gamma p + \omega_0^2 = 0$$

$$x = B e^{pt}$$

$$\frac{dx}{dt} = B p e^{pt}$$

$$\frac{d^2x}{dt^2} = B p^2 e^{pt}$$

compare $p^2 + \gamma p + \omega_0^2 = 0$

$$ax^2 + bx + c = 0$$

solution: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$p = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

$$= -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2 - 4\omega_0^2}{4}}$$

Case: 1 $\omega_0^2 > \frac{\gamma^2}{4}$

$$\omega_0^2 > \frac{\gamma^2}{4} : \text{Osc}$$

$$\omega_0^2 = \frac{\gamma^2}{4} : \text{critical}$$

$$\omega_0^2 < \frac{\gamma^2}{4} : \text{over}$$

$$\sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$= \sqrt{-\left(\omega_0^2 - \frac{\gamma^2}{4}\right)} \omega_1^2$$

$$= \sqrt{-\omega_1^2}$$

$$= \sqrt{\omega_1^2 j^2} \quad [\because j^2 = -1]$$

$$= \pm \omega_1 j$$

$$p = -\frac{\gamma}{2} \pm \omega_1 j ; p_1 = -\frac{\gamma}{2} + \omega_1 j , p_2 = -\frac{\gamma}{2} - j\omega_1$$

$$x = B_1 e^{p_1 t} + B_2 e^{p_2 t}$$

$$= B_1 e^{(-\frac{\gamma}{2} + j\omega_1)t} + B_2 e^{(-\frac{\gamma}{2} - j\omega_1)t}$$

$$= B_1 e^{-\frac{\gamma t}{2}} e^{j\omega_1 t}$$

$$= B_1 e^{-\frac{\gamma}{2}t} \cdot e^{j\omega_1 t} + B_2 e^{-\frac{\gamma t}{2}} \cdot e^{-j\omega_1 t}$$

$$= e^{-\frac{\gamma t}{2}} (B_1 e^{j\omega_1 t} + B_2 e^{-j\omega_1 t})$$

$$= A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

$$y = 5 e^{-3x} \cos(3x)$$

$$\rho = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$\rho = -\frac{\gamma}{2}, \quad x = (A + Bt) e^{-\frac{\gamma}{2}t}, \quad y = 2e^{-5x}$$

Physics, class: 10

$$\textcircled{1} \text{ Osci } \omega^2 > \frac{\gamma^2}{4}$$

$$\textcircled{2} \text{ critical } \omega^2 = \frac{\gamma^2}{4}$$

$$\textcircled{3} \text{ Overdamp } \omega^2 < \frac{\gamma^2}{4}$$

$$\rho_1 = -\frac{\gamma}{2} + \lambda, \quad \rho_2 = -\frac{\gamma}{2} - \lambda$$

$$x = B_1 e^{(-\frac{\gamma}{2} + \lambda)t} + B_2 e^{(-\frac{\gamma}{2} - \lambda)t}$$

$$\text{critical } x(t) = (A + Bt) e^{-\frac{\gamma}{2}t}$$

$$x(t) = A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \phi)$$

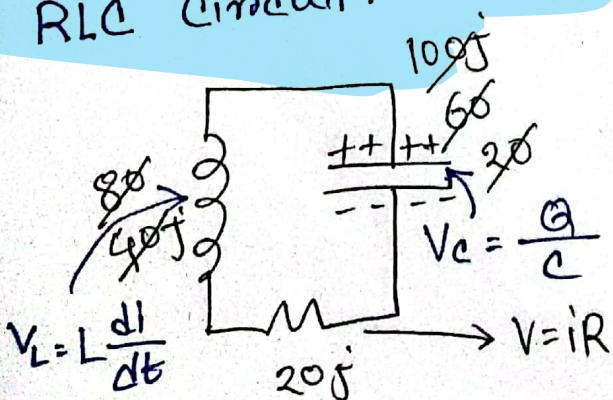
$$x = B e^{\rho t}$$

$$\rho = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

RLC circuit:

some energy will be captured by resistor.

discharged to charged.



$$iR = L \frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} \cdot Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0$$

RLC

$$\gamma = \frac{R}{L}$$

$$\omega_0^2 = \frac{1}{LC}$$

mass-spring

$$f = \frac{1}{m}, \omega_0^2 = \frac{k}{m}$$

Mass-spring

LC circuit

RC circuit

$$\frac{k}{m} > \frac{\gamma^2}{4m^2}$$

$$\omega_0^2 > \frac{\gamma^2}{4} \text{ Osci}$$

$$\frac{1}{LC} > \frac{R^2}{4L^2}$$

$$\frac{k}{m} = \frac{\gamma^2}{4m^2}$$

$$\omega_0^2 = \frac{\gamma^2}{4} \text{ critical}$$

$$\frac{1}{LC} = \frac{R^2}{4L^2}$$

$$\frac{k}{m} < \frac{\gamma^2}{4m^2}$$

$$\omega_0^2 < \frac{\gamma^2}{4} \text{ over}$$

$$\frac{1}{LC} < \frac{R^2}{4L^2}$$

case-1: $\frac{1}{LC} > \frac{R^2}{4L^2}$

$$Q(t) = Ae^{-\frac{\gamma t}{2}} \cos(\omega_0 t + \phi)$$

$$= Ae^{-\frac{Rt}{2L}} \cos(\omega_0 t + \phi)$$

damped angular frequency

$$x = A \cos(\omega t), \quad \omega_1^2 = \omega_0^2 - \frac{\gamma^2}{4}, \quad \omega_1^2 = \sqrt{\left(\frac{1}{LC}\right) - \frac{R^2}{4L^2}}$$

Ex: 1 $C = 1 \mu F = 10^{-6} F$

$L = 0.2 H$ $h = \text{henri}$

$R = 800 \Omega$

$$\frac{1}{LC} = \frac{1}{0.2 \times 10^{-6}} = 5 \times 10^6$$

$$\frac{R^2}{4L^2} = \frac{(800)^2}{4(0.2)^2} = 4 \times 10^6$$

different pattern

$m = 0.5 \text{ kg}$ $\omega^2 = \frac{k}{m}$

$k = 196 \text{ N/m}$

$\frac{\gamma^2}{4} = \frac{b^2}{4m^2}$

$b = 25$

$$\omega_1 = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$2\pi f_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

$$= \frac{1}{2\pi} \sqrt{(5 \times 10^6) - (4 \times 10^6)}$$

15.06

undamped $\omega^2 = \left(\frac{2\pi}{T}\right)^2$ $T = 2\pi \sqrt{\frac{m}{k}}$

SHM $\omega_0^2 = \frac{k}{m}$

DHM $\omega_0^2 = \sqrt{\frac{k}{m} - \left(\frac{b^2}{4m^2}\right)}$ we can ignore $= \frac{k}{m}$

$\downarrow$
 $x(t) = Ae^{-\frac{\gamma t}{2}}$

$\left(\frac{A}{2}\right) = Ae^{-\frac{bt}{2m}}$

$\Rightarrow \frac{1}{2} = e^{-\frac{bt}{2m}}$

$\Rightarrow \ln\left(\frac{1}{2}\right) = -\frac{bt}{2m} (\ln e) \rightarrow 1$

$\Rightarrow t = -\frac{2m}{b} \ln\left(\frac{1}{2}\right)$

$$E_p = \frac{1}{2} k (A)^2$$

$$= \frac{1}{2} k \left(A e^{-\frac{bt}{2m}} \right)^2$$

$$= \frac{1}{2} k A^2 e^{-\frac{bt}{m}}$$

initial $\frac{1}{2} k A^2$

$$= \frac{1}{2} k \left(A e^{-\frac{bt}{2m}} \right)^2 = \frac{1}{2} (k A^2)$$

$$= \left(\frac{1}{2} k A^2 \right) e^{-\frac{bt}{m}} = \frac{1}{2} (k A^2)$$

$$\Rightarrow e^{-\frac{bt}{m}} = \frac{1}{2}$$

$$\Rightarrow -\frac{bt}{m} = \ln \frac{1}{2}$$

$$\Rightarrow t = -\frac{b}{m} \ln \left(\frac{1}{2} \right)$$